

Content Refresh Prioritization using Search Intelligence

An Explainable Machine Learning Approach for Ranking Content Update Opportunities

Muhammad Naveed Qasim · Bachelor of Artificial Intelligence · Riphah International University · FlyRank ML Internship

Abstract

Content teams cannot manually review thousands of pages to decide which need refreshing. This research frames content refresh prioritization as a binary classification problem — declining vs. not declining — and compares a transparent rule-based baseline against a Random Forest model trained on 12 search-and-content features derived from the FlyRank ML Internship dataset: 79 million-plus anonymized production search records, stored in DuckDB and derived from Google Search Console signals. The model improves precision and recall over the baseline while remaining fully explainable via feature importance, and its output is a ranked, reason-coded worklist intended purely as decision support for a human editorial team.

Research Question

How can search intelligence signals identify declining content pages that should be refreshed before significant traffic loss occurs?

Method

Data cleaning → feature engineering (12 features) → rule-based baseline → Random Forest classifier → GroupShuffleSplit validation with leakage checks → evaluation against the base rate → conversion of per-page probabilities into High / Medium / Low priority tiers with reason codes.

Features

content_age_days, days_since_last_update, impressions_90d, avg_position, ctr, word_count, visible_queries, rare_share, anon_share, top_query_share, imp_prev30, imp_last30.

Results (placeholder figures, pending final sign-off)

Metric	Baseline rule	Random Forest
Accuracy	0.62	0.81
Precision	0.41	0.67
Recall	0.58	0.71
F1 score	0.48	0.69

Base rate	0.18	0.18
-----------	------	------

Recommendation Reason Codes

STALE_VISIBLE — no update in 12+ months, still ranking. **DECLINING_SIGNAL** — impressions down vs. the prior period. **HIGH_EXPOSURE** — still among the page's top exposure tier. **LOW_PRIORITY** — stable or improving trend.

Limitations

Observed relationships only, not proven causes. Decision support only — no autonomous content actions. No causal claims. Does not predict or reverse-engineer Google's ranking algorithm. No client-identifying information is used anywhere in this research.

Built on the FlyRank ML Internship dataset (flyrank.ai) · github.com/Naveed-Qasim608/Flyrank_ML_Internship · linkedin.com/in/muhammad-naveed-qasim-7760573a8